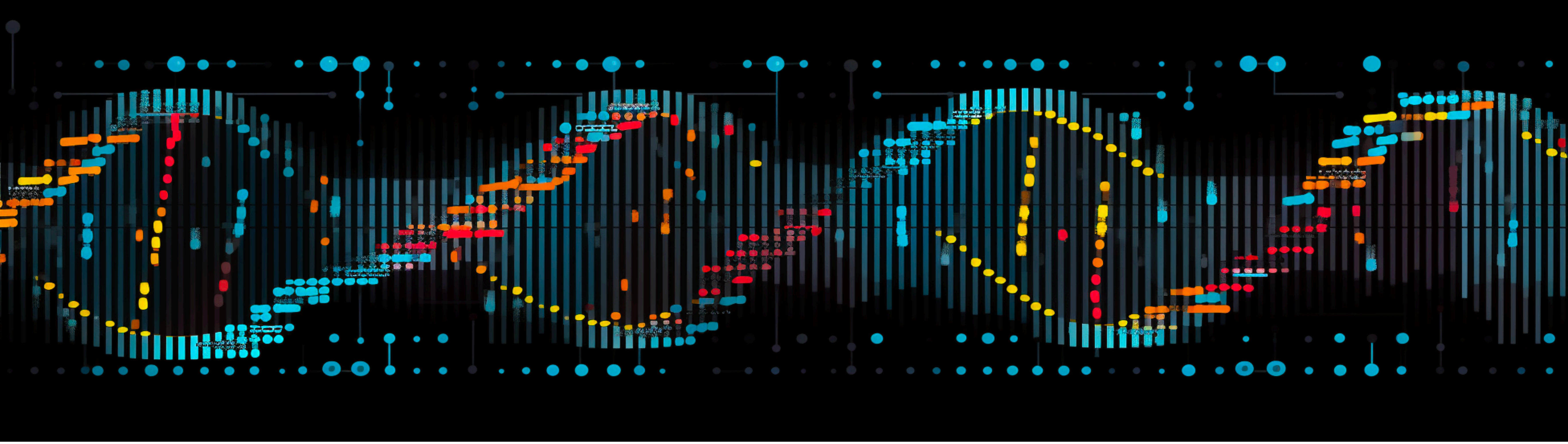
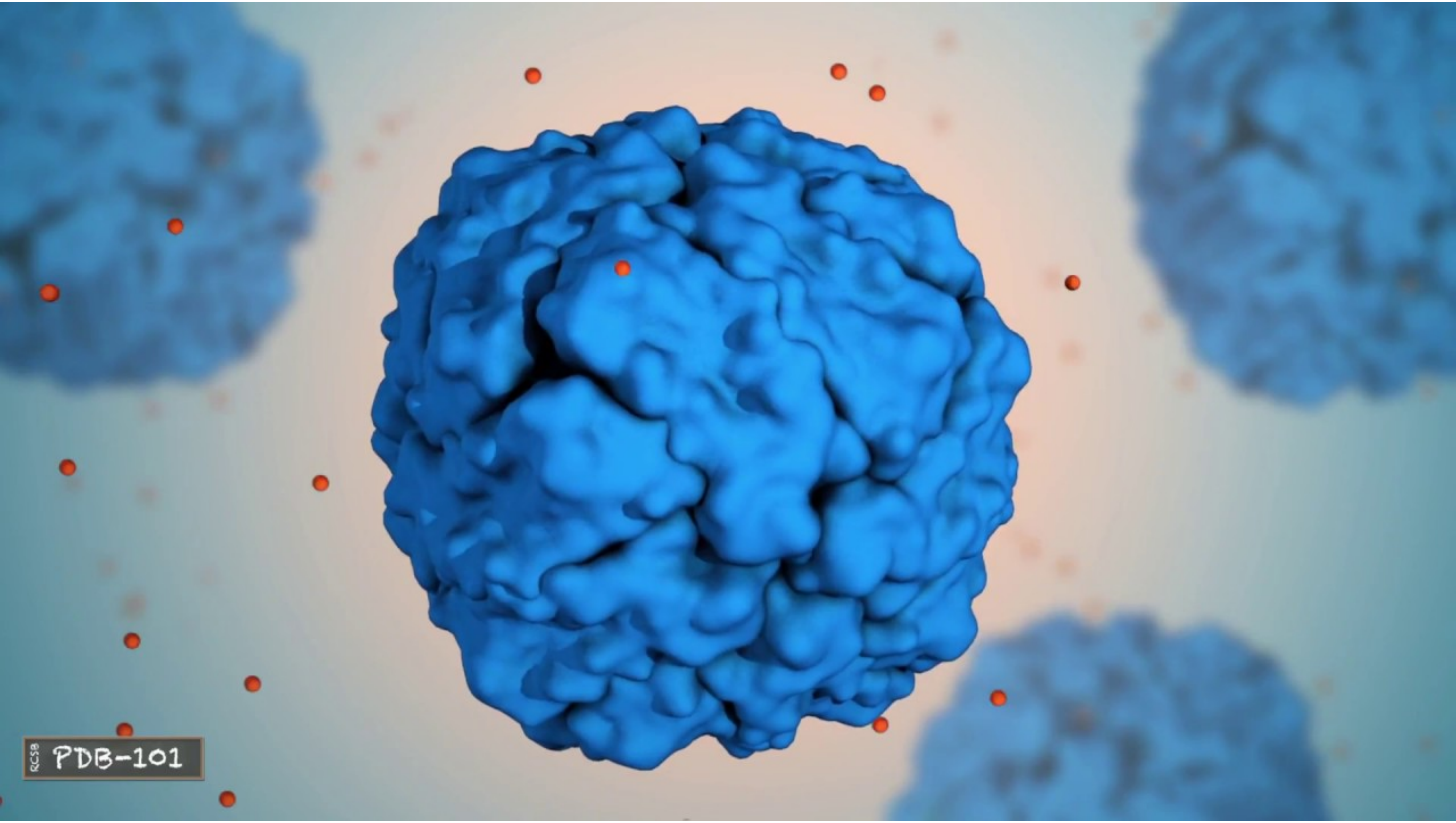


Introduction to bioinformatics

Chapter2 part4 Protein and CRISPR

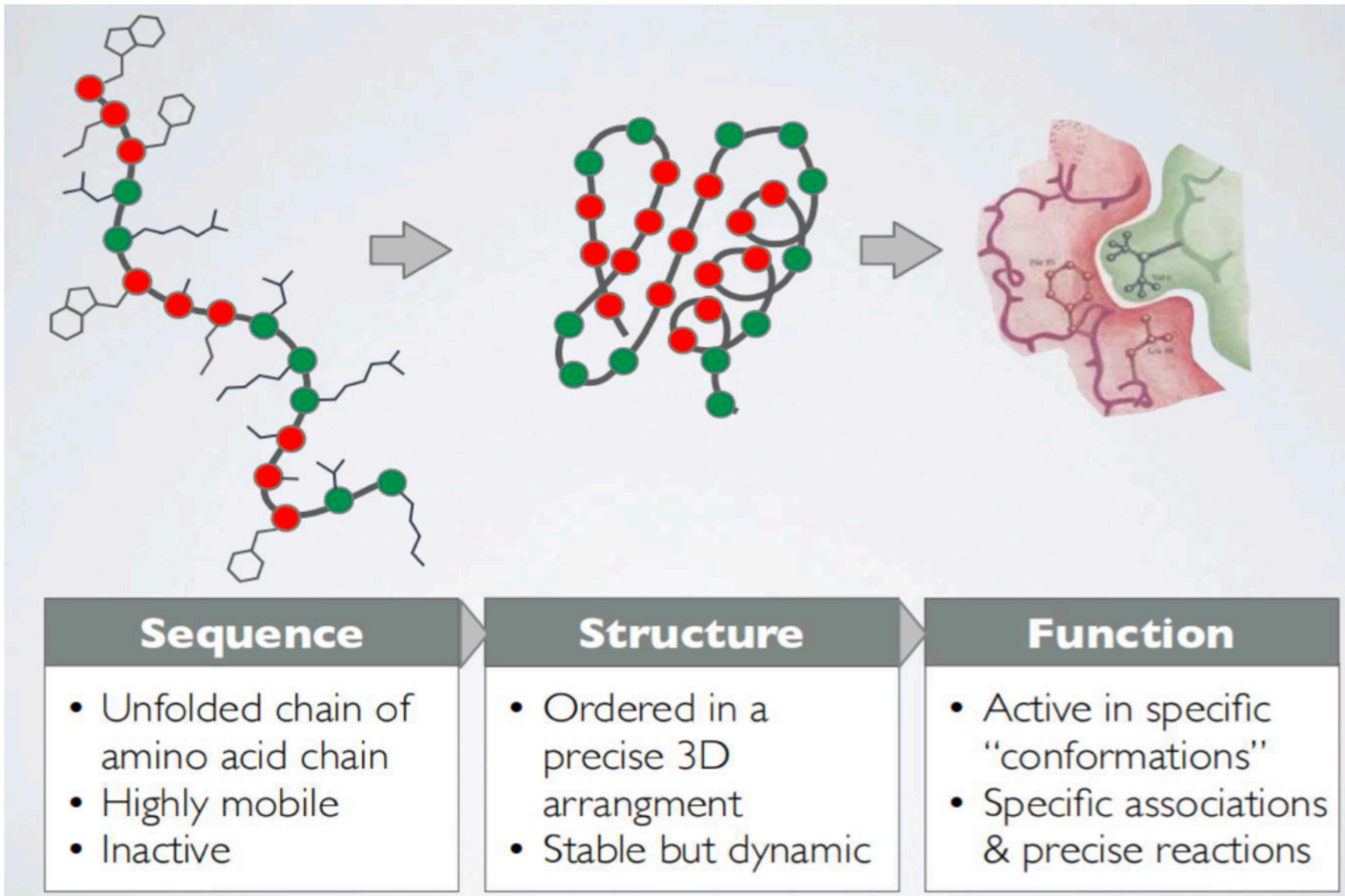
Jiaxing Chen

What is a Protein?

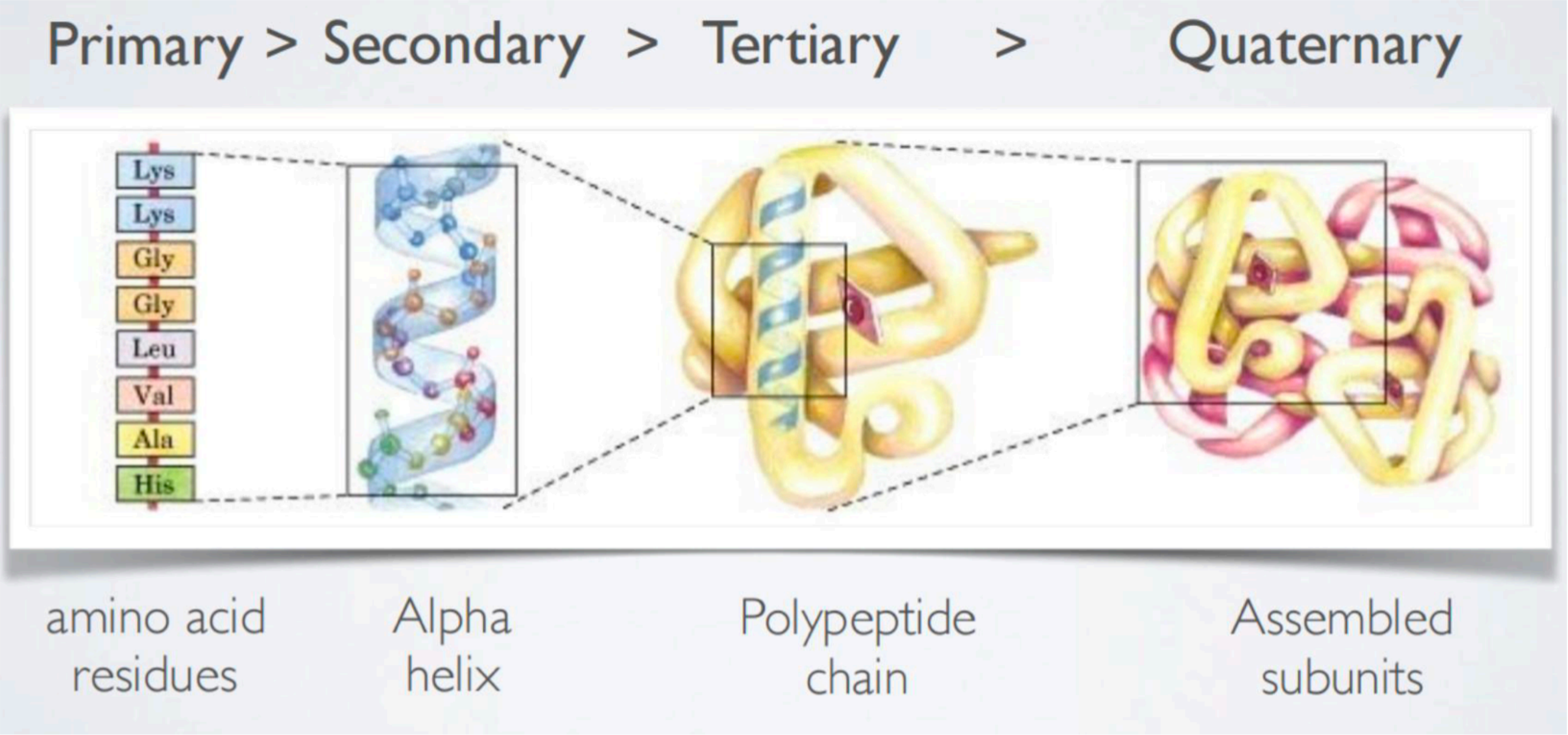


https://www.youtube.com/watch?v=wvTv8TqWC48

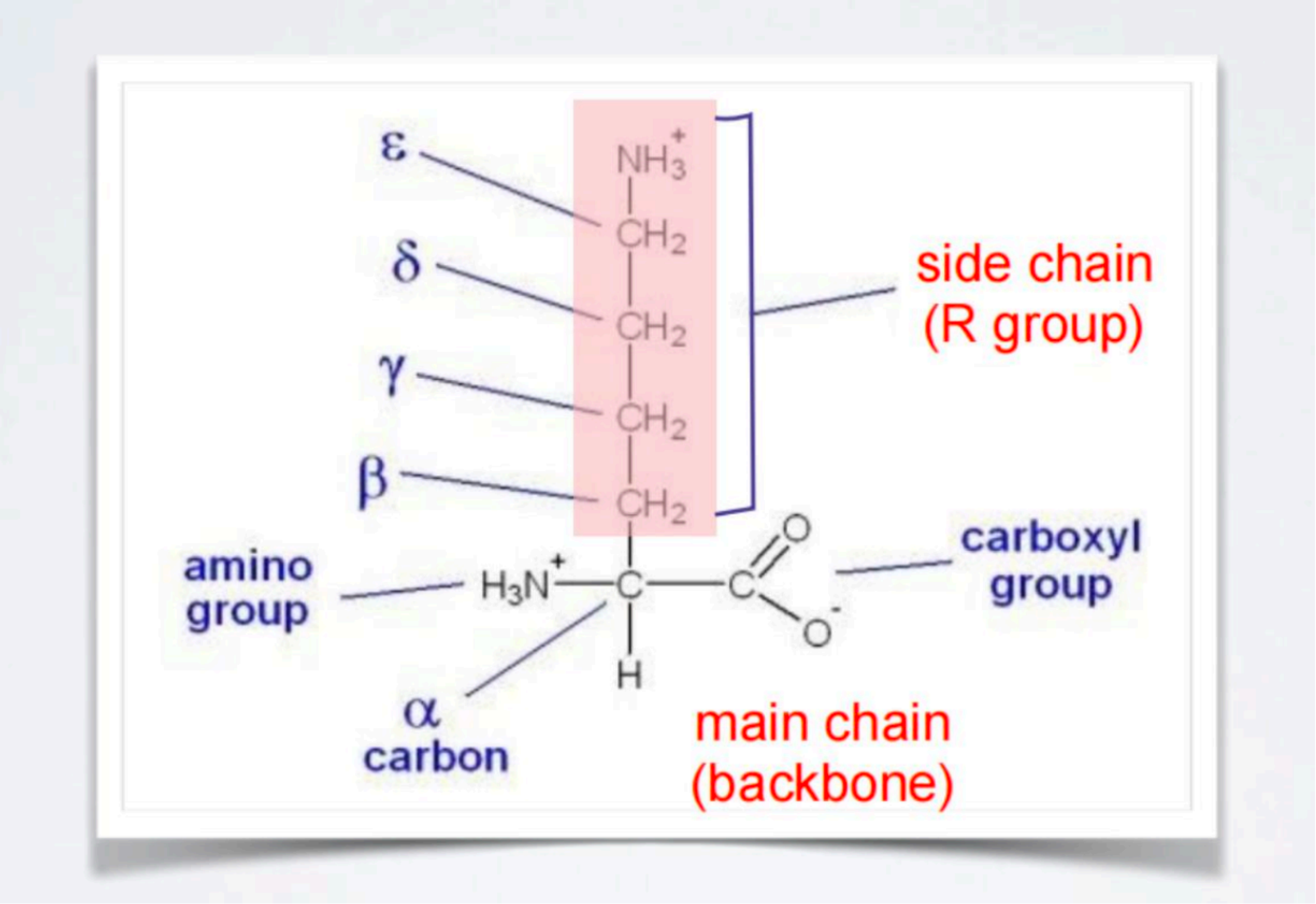
STRUCTURAL DATA IS CENTRAL



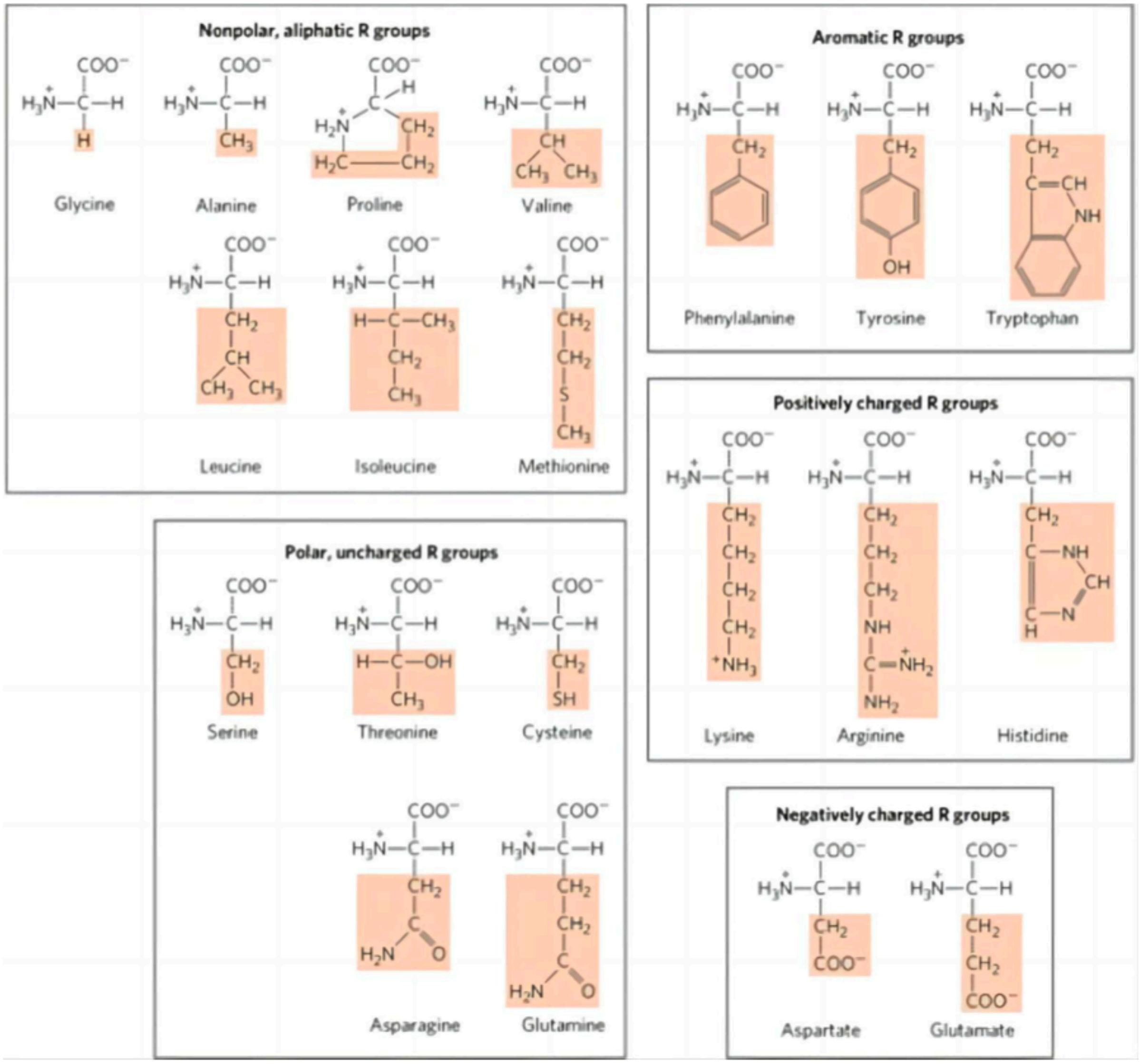
HIERARCHICAL STRUCTURE OF PROTEINS



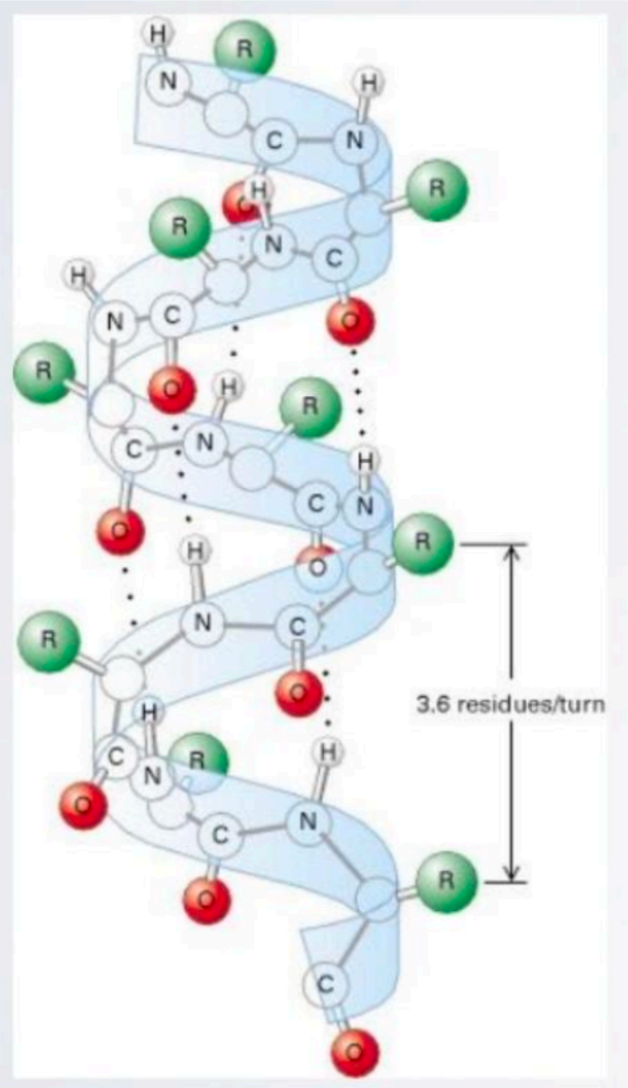
RECAP: AMINO ACID NOMENCLATURE



AMINO ACIDS CAN BE GROUPED BY THE PHYSIOCHEMICAL PROPERTIES



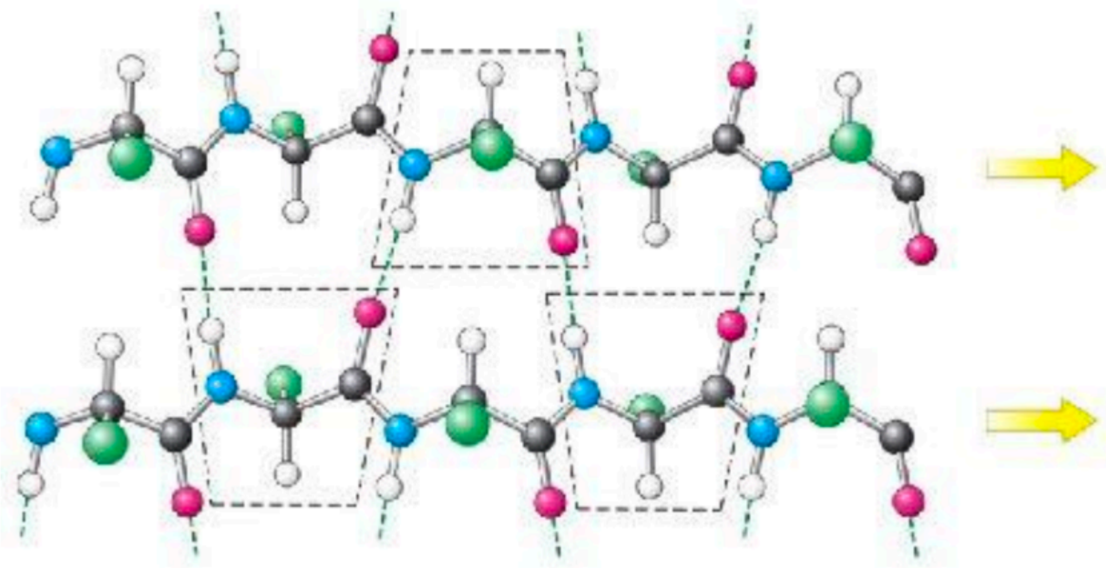
MAJOR SECONDARY STRUCTURE TYPES
ALPHA HELIX & BETA SHEET



α-helix

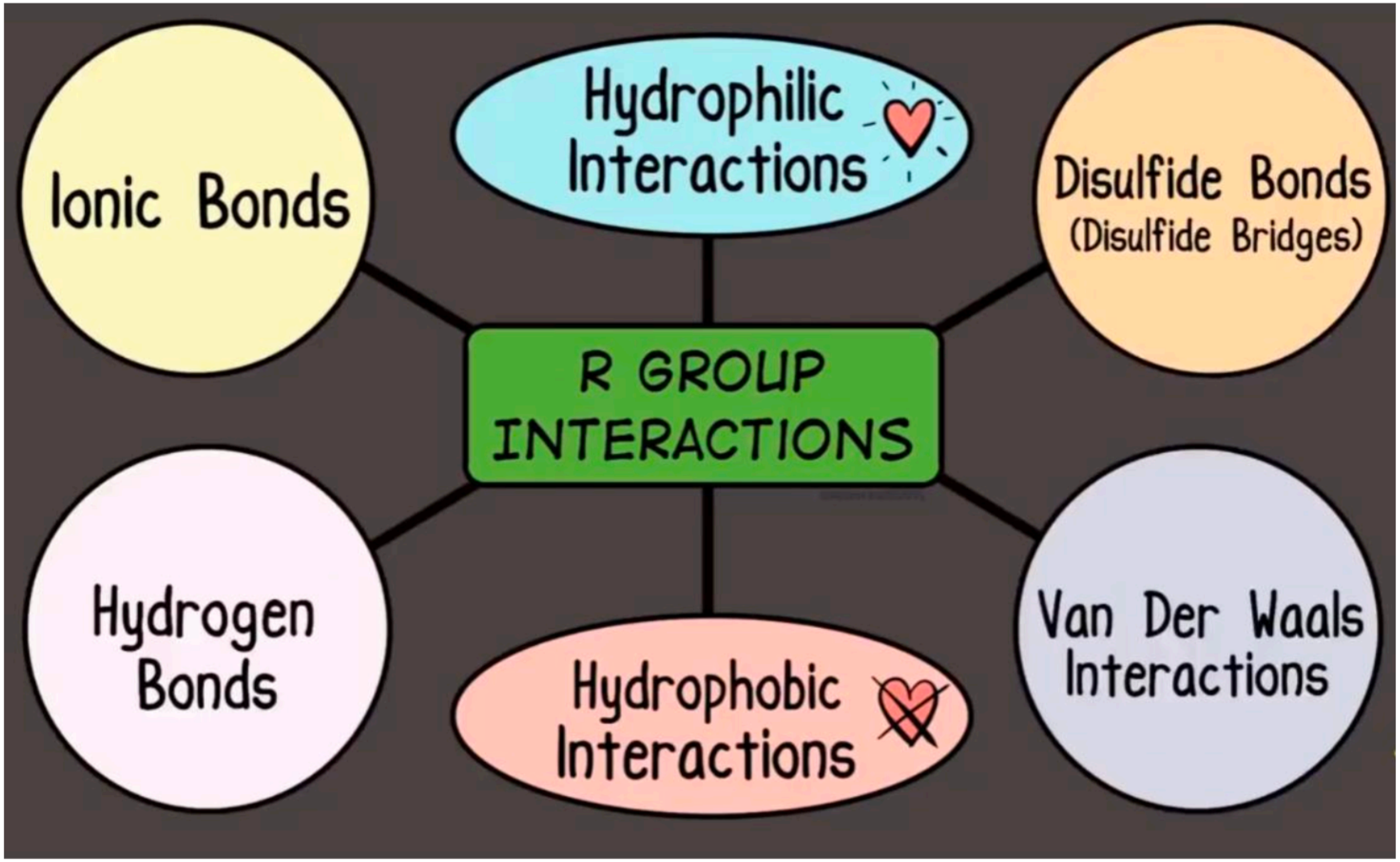
• Most common from has 3.6 residues per turn (number of residues in one full rotation)
• Hydrogen bonds (dashed lines) between residue i and i+4 stabilize the structure
• The side chains (in green) protrude outward
• 3_{10}-helix and π-helix forms are less common

MAJOR SECONDARY STRUCTURE TYPES
ALPHA HELIX & **BETA SHEET**

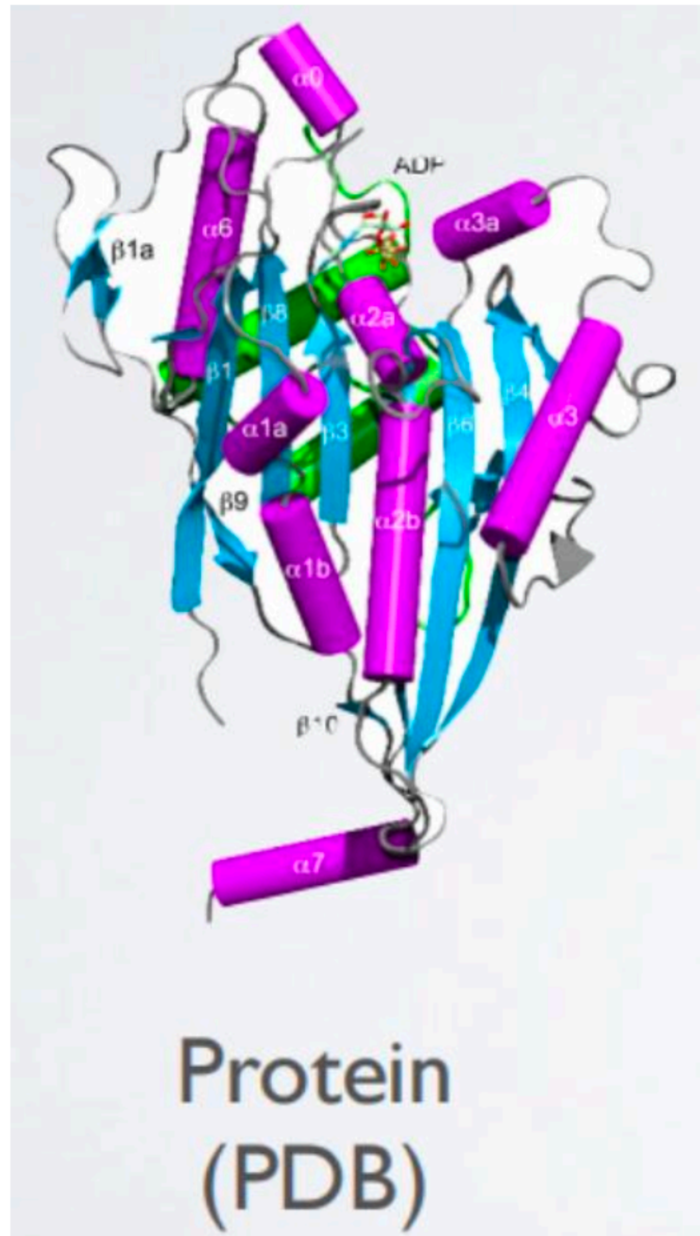


In parallel β-sheets
- Adjacent β-strands run in same direction
- Hydrogen bonds (dashed lines) between NH and CO stabilize the structure
- The side chains (in green) are above and below the sheet

Key forces affecting Tertiary Structure

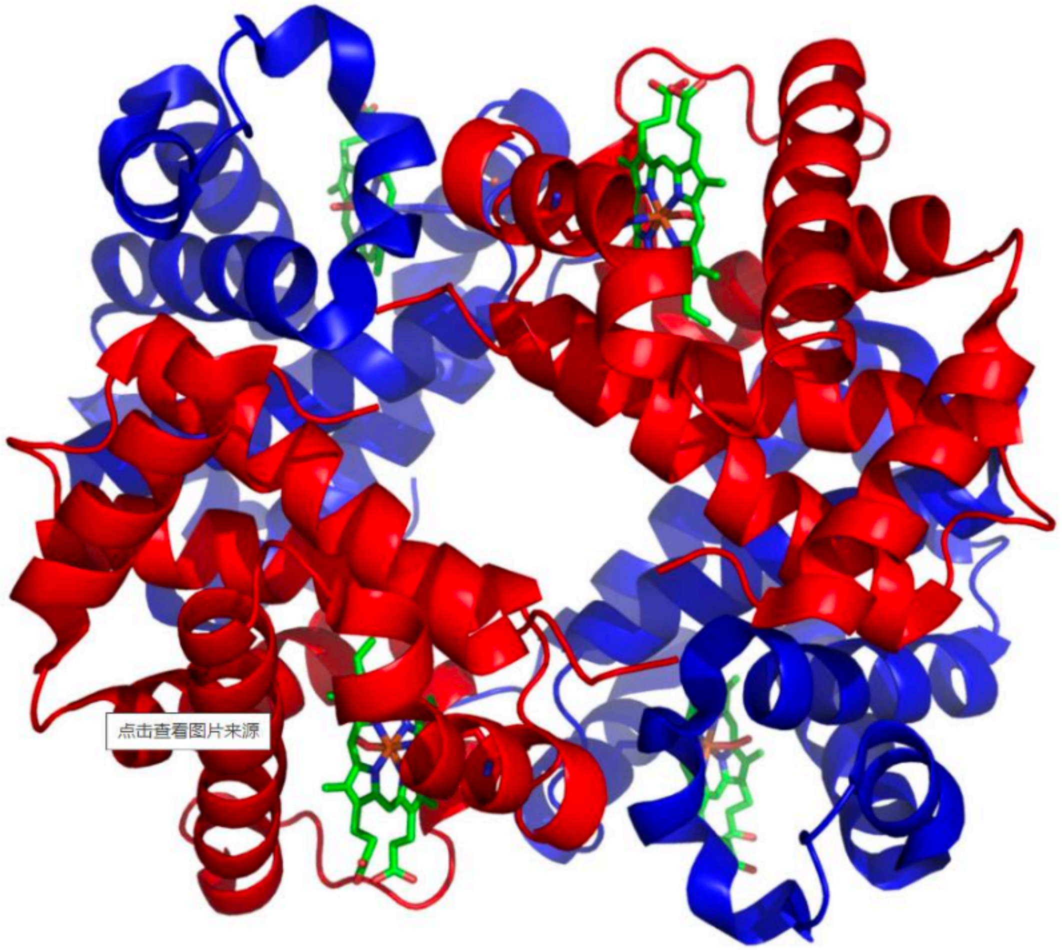


Tertiary Structure of an example protein



Quaternary Structure

- Some proteins consists of multiple peptide chains



Hemoglobin

Protein domains and motifs: Definitions

Signature:

- a protein category such as a domain or motif

Domain:

- a region of a protein that can adopt a 3D structure
- a fold
- a family is a group of proteins that share a domain

examples: zinc finger domain; immunoglobulin domain

Motif (or fingerprint):

- a short, conserved region of a protein
- typically 10 to 20 contiguous amino acid residues

15 most common domains (human)

Zn finger, C2H2 type	1093 proteins
Immunoglobulin	1032
EGF-like	471
Zn-finger, RING	458
Homeobox	417
Pleckstrin-like	405
RNA-binding region RNP-1	400
SH3	394
Calcium-binding EF-hand	392
Fibronectin, type III	300
PDZ/DHR/GLGF	280
Small GTP-binding protein	261
BTB/POZ	236
bHLH	226
Cadherin	226

Definition of a domain

According to InterPro at EBI (http://www.ebi.ac.uk/interpro/):

- A domain is an independent structural unit, found alone or in conjunction with other domains or repeats.
- Domains are evolutionarily related.

According to SMART (http://smart.embl-heidelberg.de):

- A domain is a conserved structural entity with distinctive secondary structure content and a hydrophobic core.
- Homologous domains with common functions usually show sequence similarities.

Definition of a motif

A motif (or fingerprint) is a short, conserved region of a protein. Its size is often 10 to 20 amino acids.

Simple motifs include transmembrane domains and phosphorylation sites. These do not imply homology when found in a group of proteins.

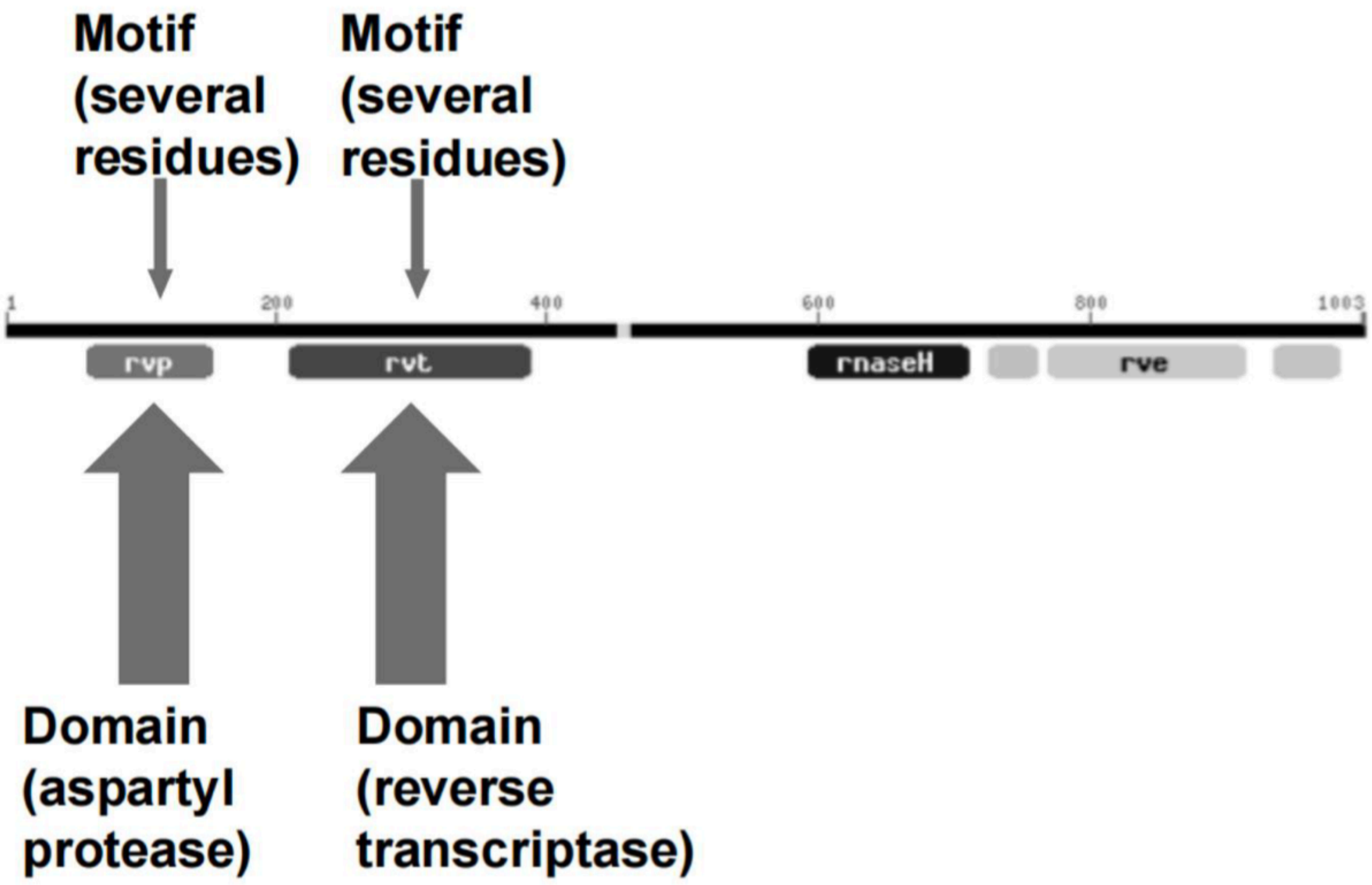
PROSITE (www.expasy.org/prosite) is a dictionary of motifs (there are currently 1600 entries). In PROSITE, a pattern is a qualitative motif description (a protein either matches a pattern, or not). In contrast, a profile is a quantitative motif description. We will encounter profiles in Pfam, ProDom, SMART, and other databases.

Protein domain vs motif

A structural domain is an element of the protein's overall structure that is stable and often folds independently of the rest of the protein chain. Like the PH domain above, many domains are not unique to the protein products of one gene, but instead appear in a variety of proteins. Proteins sharing more than a few common domains are encoded by members of evolutionarily related genes comprising **gene families**.

Protein motifs are small regions of protein three-dimensional structure or amino acid sequence shared among different proteins. They are recognizable regions of protein structure that may (or may not) be defined by a unique chemical or biological function.

Proteins can have both domains and motifs (patterns)



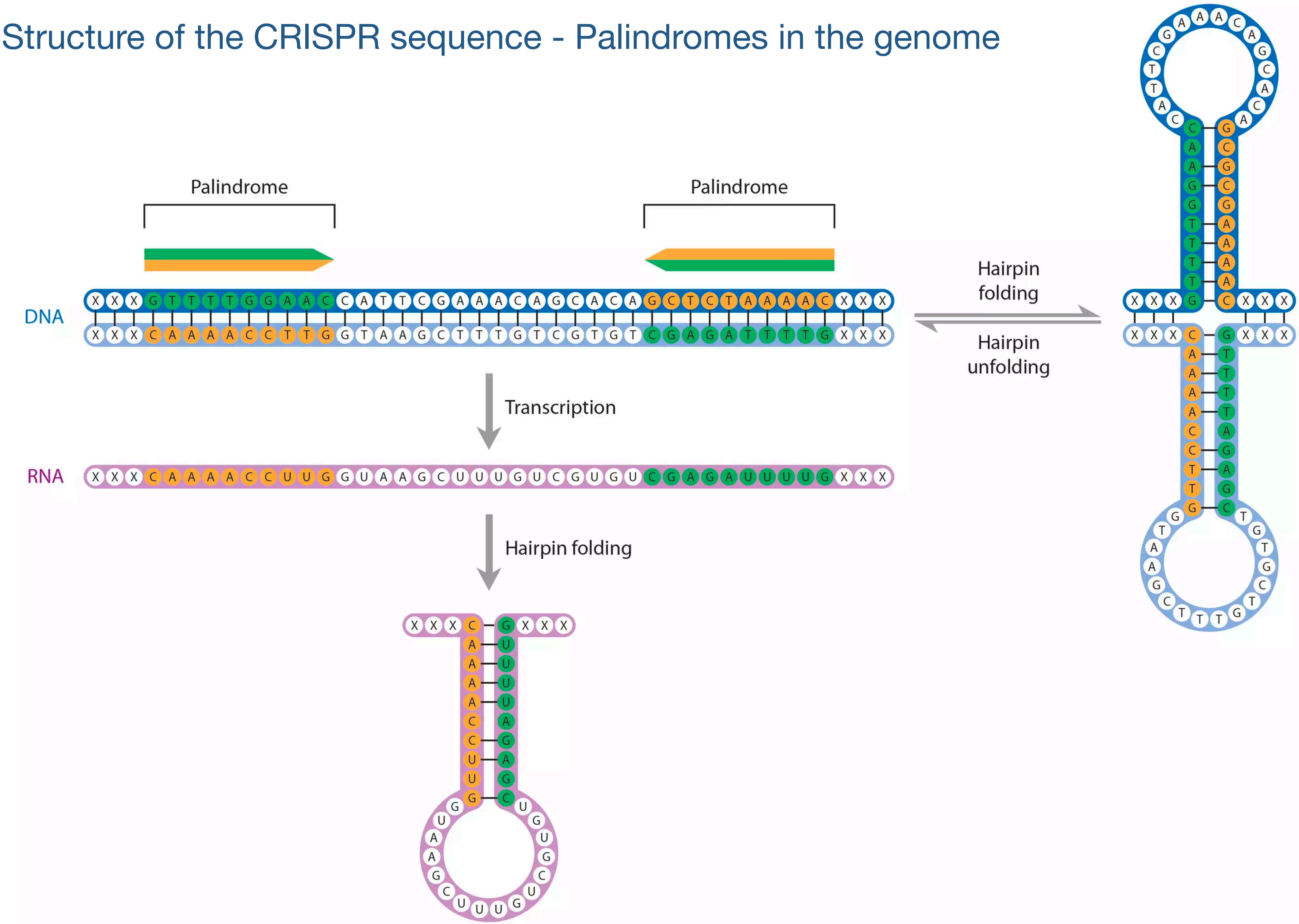
A true story on protein mining

Once upon a time, scientists found a mysterious region in bacterial genome

looks like a diamond necklace

They did not know the function of this region back then,
so they just named it as CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats)

Structure of the CRISPR sequence - Palindromes in the genome

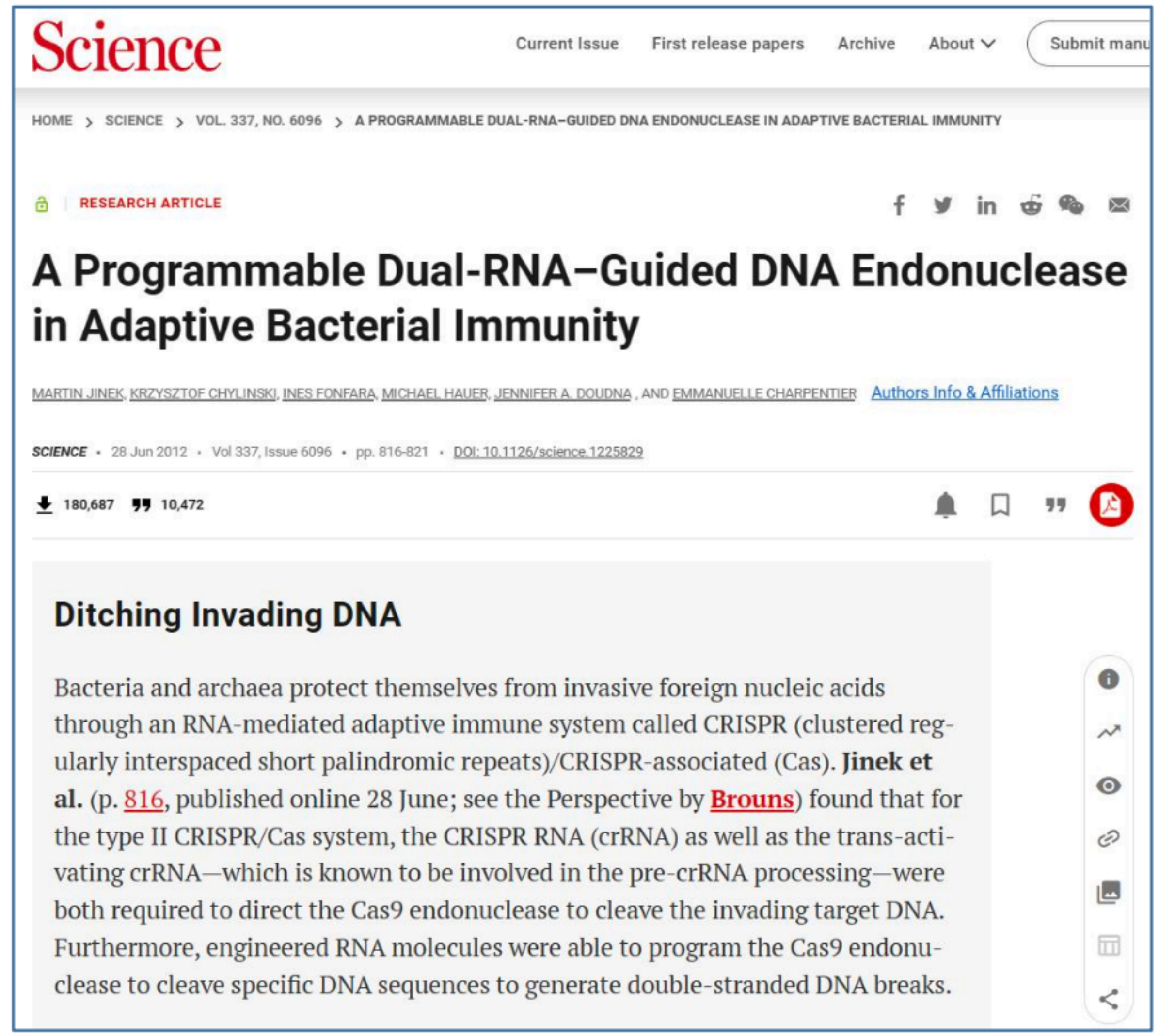


About 40 years later, in 2012, two female scientists broke the codes

埃玛纽埃尔·卡彭蒂耶
Emmanulle Charpentier

詹妮弗·杜德娜
Jennifer Doudna

Nobel Prize Winners in 2020



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RESEARCH ARTICLE

A Programmable Dual-RNA–Guided DNA Endonuclease in Adaptive Bacterial Immunity

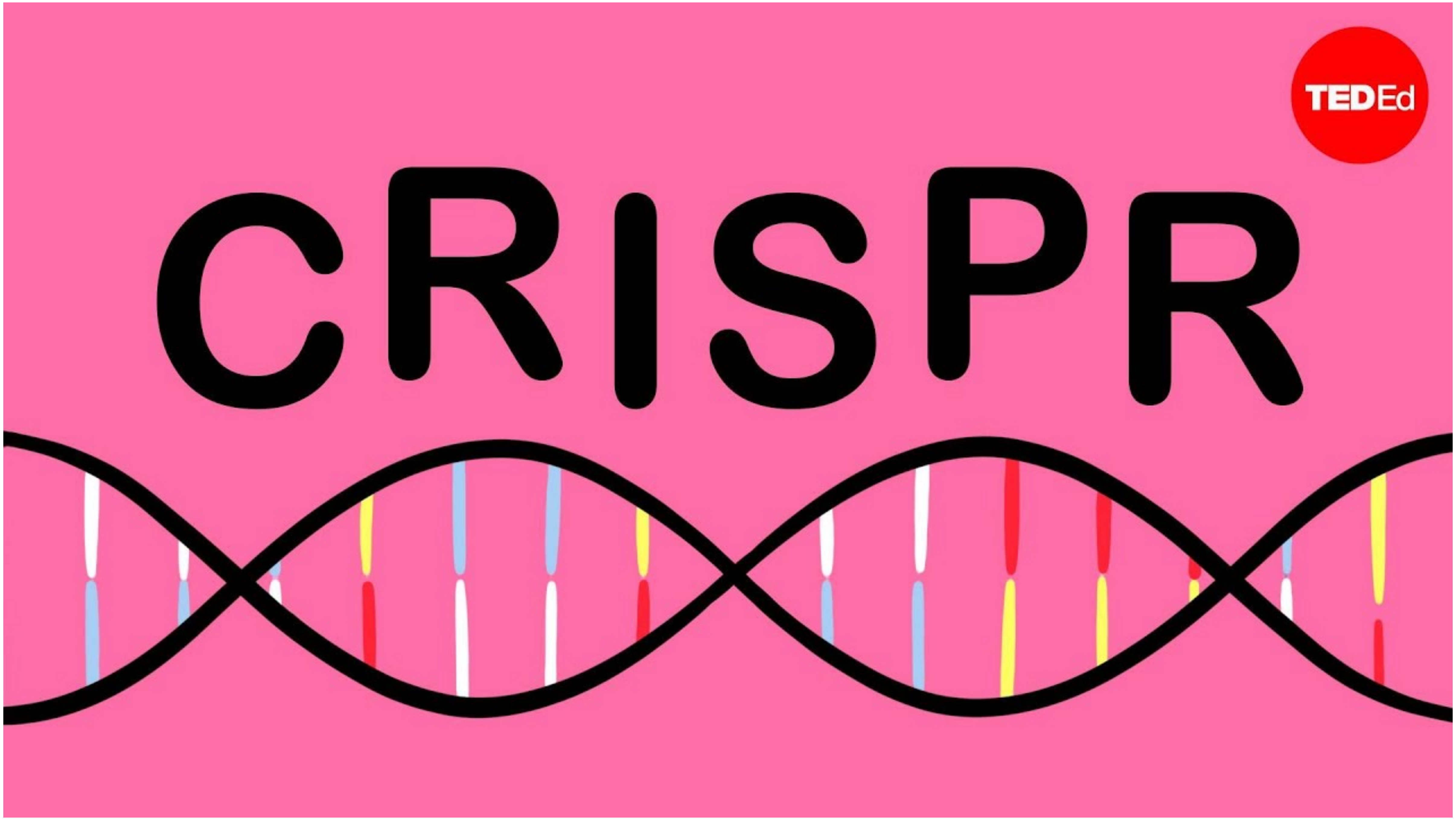
MARTIN JINEK, KRZYSZTOF CHYLINSKI, INES FONFARA, MICHAEL HAUER, JENNIFER A. DOUDNA , AND EMMANUELLE CHARPENTIER Authors Info & Affiliations

SCIENCE • 28 Jun 2012 • Vol 337, Issue 6096 • pp. 816-821 • DOI: 10.1126/science.1225829

180,687 10,472

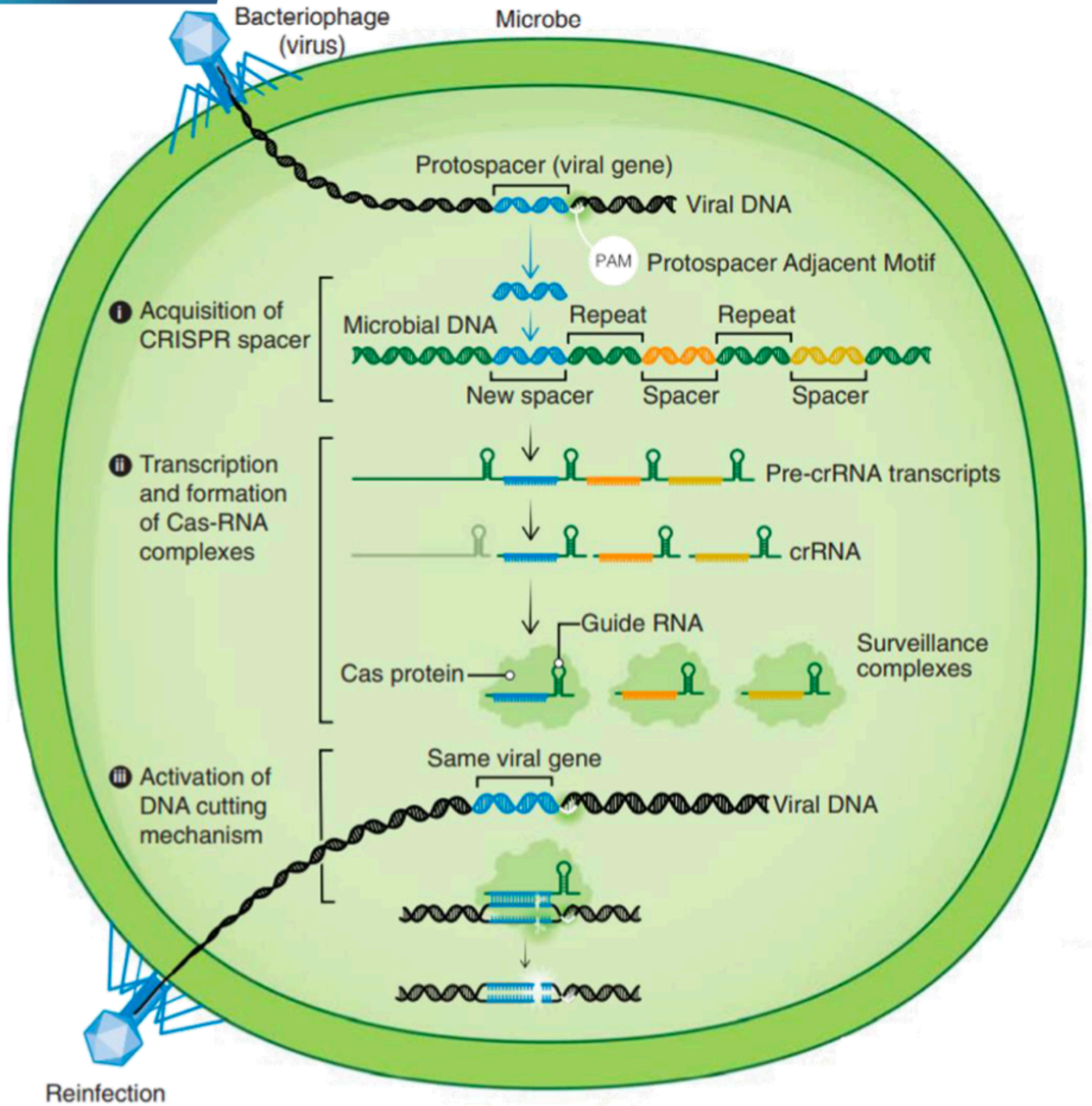
Ditching Invading DNA

Bacteria and archaea protect themselves from invasive foreign nucleic acids through an RNA-mediated adaptive immune system called CRISPR (clustered regularly interspaced short palindromic repeats)/CRISPR-associated (Cas). **Jinek et al.** (p. 816, published online 28 June; see the Perspective by **Brouns**) found that for the type II CRISPR/Cas system, the CRISPR RNA (crRNA) as well as the trans-activating crRNA—which is known to be involved in the pre-crRNA processing—were both required to direct the Cas9 endonuclease to cleave the invading target DNA. Furthermore, engineered RNA molecules were able to program the Cas9 endonuclease to cleave specific DNA sequences to generate double-stranded DNA breaks.



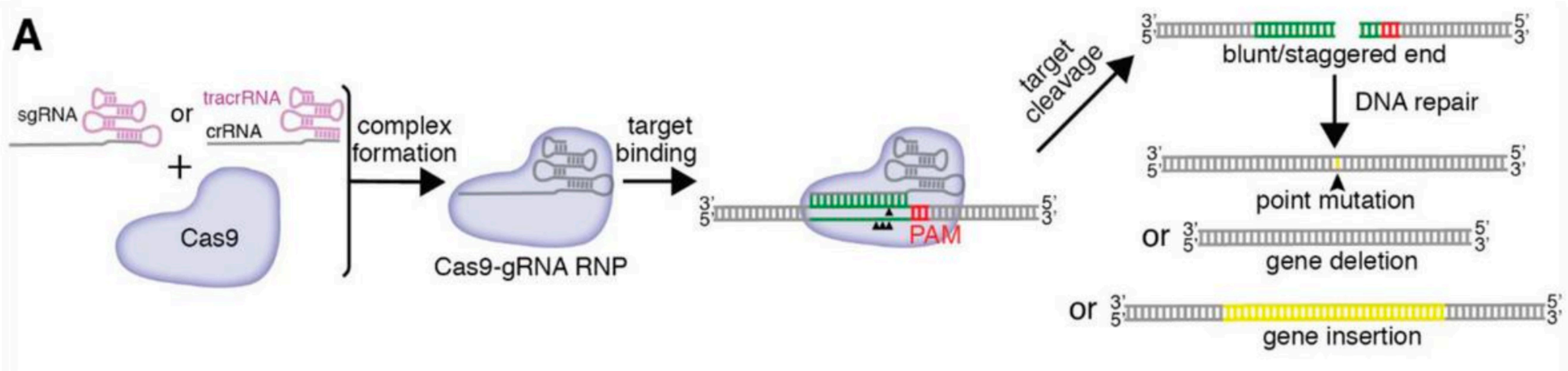
https://www.youtube.com/watch?v=6tw_JVz_IEc

CRISPR-Cas system: a defense system



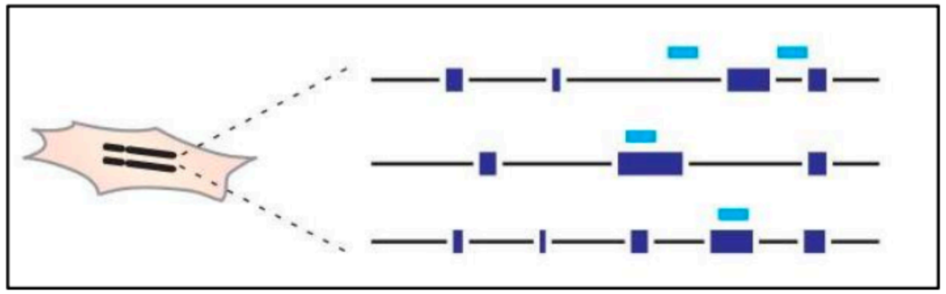
- The CRISPR-Cas system in bacteria serves as a **defense mechanism** against foreign genetic material, such as viruses and plasmids

From CRISPR-Cas systems to Genome editing tools

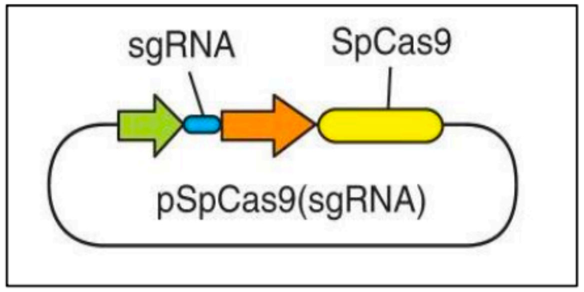


- Cas nucleases induce **DNA double-strand breaks (DSB)** at desired locations within a genome.
- DSBs are themselves highly **genotoxic lesions** and as such cells have evolved multiple mechanisms for their repair:
  - **NHEJ**: Non-homologous end joining
  - **HDR**: Homology-directed repair
- Further processing of the DSBs by the cellular DSB repair machinery is then necessary to introduce desired **mutations**, **sequence insertions**, or **gene deletions**.

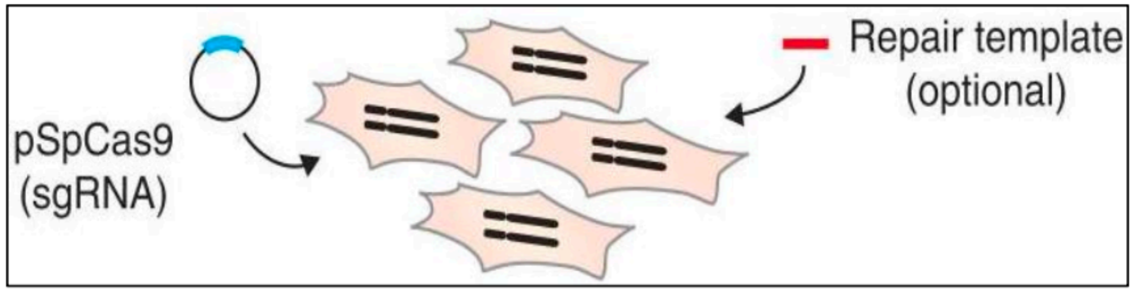
From CRISPR-Cas systems to Genome editing tools



Step 1



Step 2

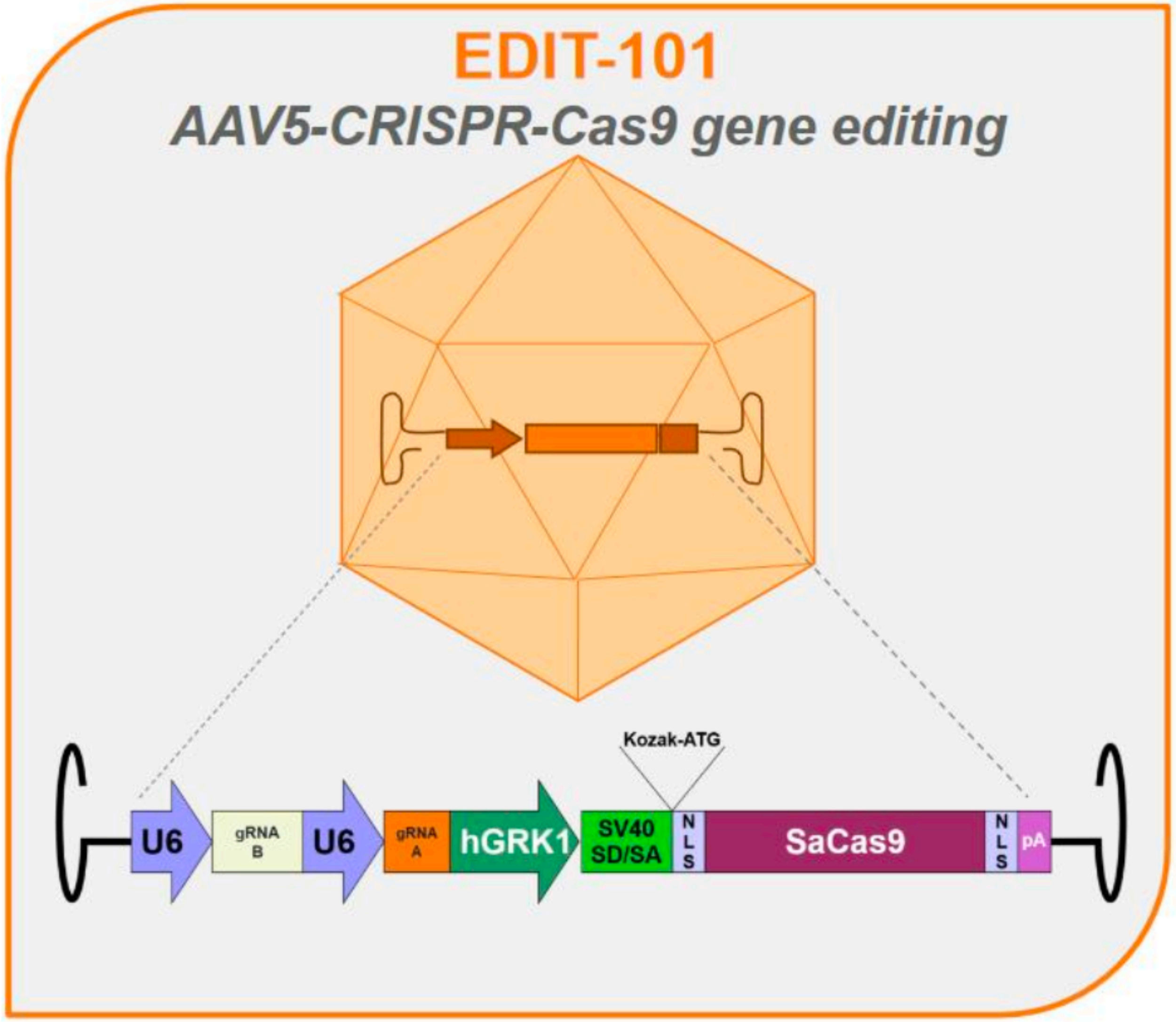


Step 3

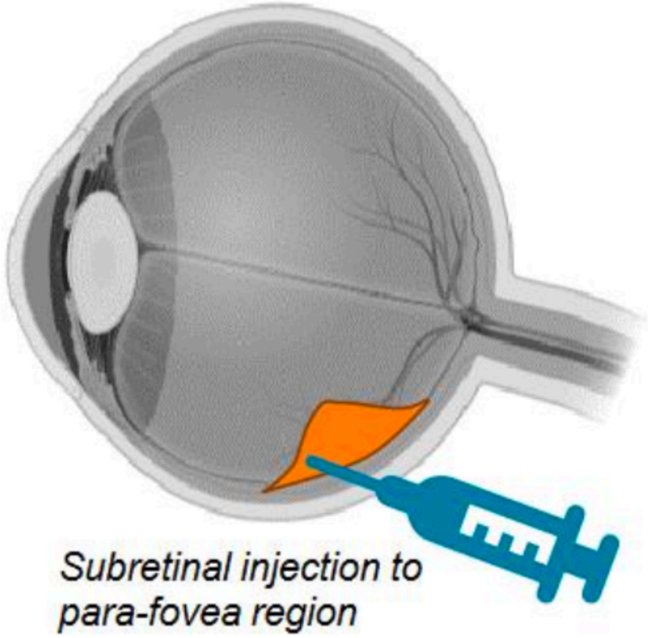
- Experimental gene editing using an **artificially designed and synthesized** Cas9 expression vector.
- Overview of a genome editing experiment:
  - Step 1. In silico target/gRNA design
  - Step 2. Expression vector construction
  - Step 3. Expression vector delivery
- CRISPR-Cas system is a **programmable** genome editing tools

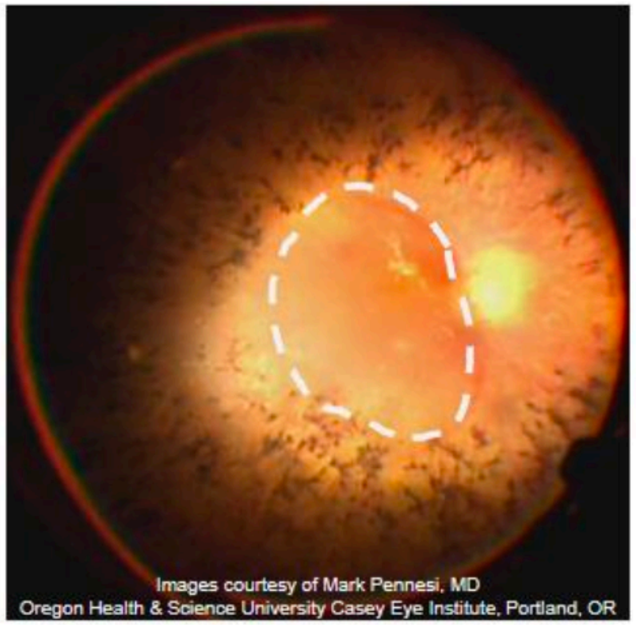
Applications of CRISPR-Cas System (I)

- EDIT-101: Gene therapy for *CEP290* gene (related to retinal degeneration)



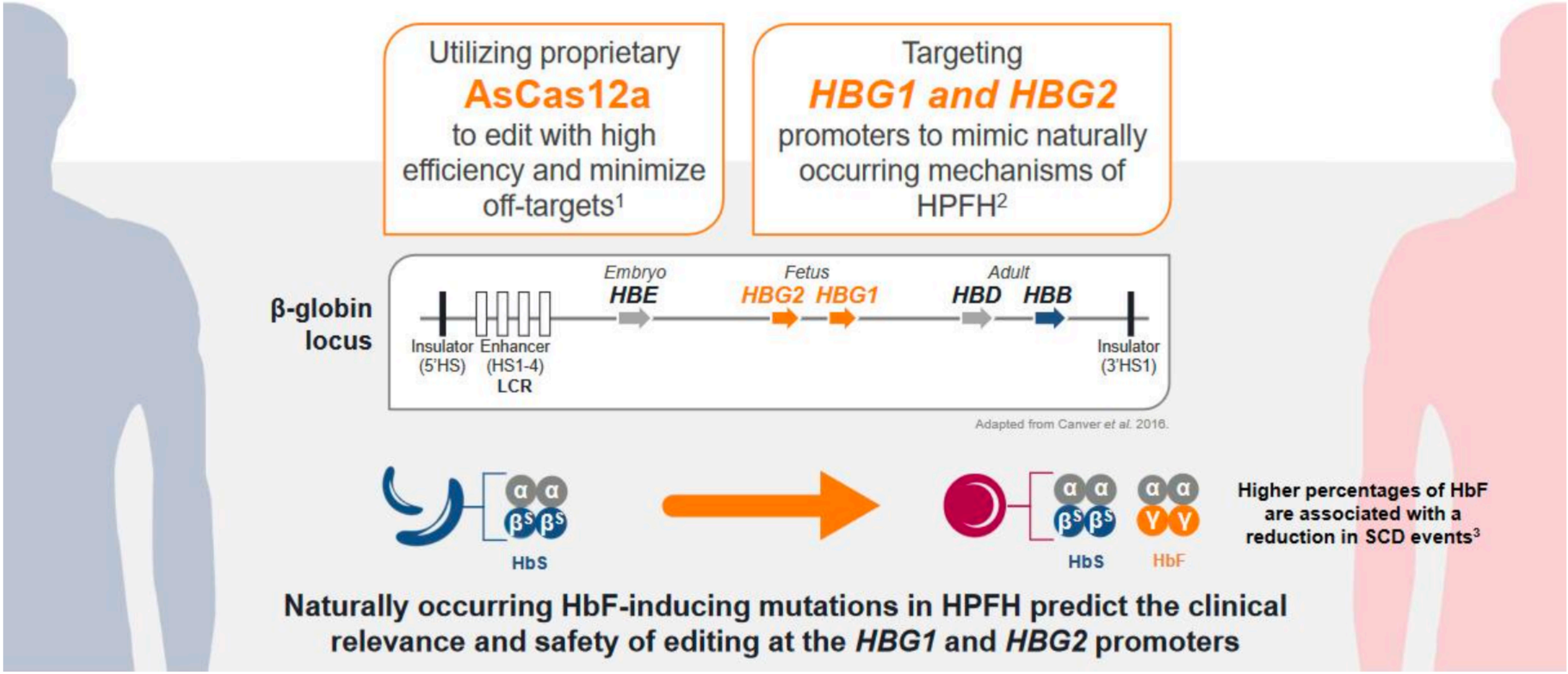
- 视网膜变性 -> 先天性黑蒙症 (*CEP290* 突变/缺失)
- AAV5 with DNA encoding 2 gRNAs and SaCas9 delivered subretinally as a single dose.





Applications of CRISPR-Cas System (II)

- EDIT-301: AsCas12a –based gene therapy for *HBG1* & *HBG2* gene (related to hemoglobin SC disease, SCD)




THANK YOU